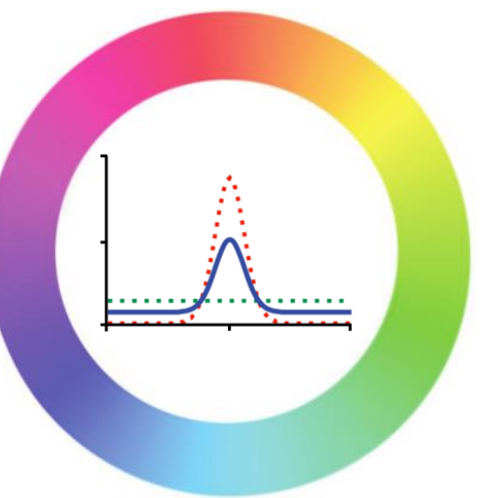




Competition Between Working Memory Precision and Hippocampal Pattern Separation

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INTRODUCTION

✓ Working Memory Precision

Visual WM consists of capacity (quantitative) and precision (qualitative)

✓ Hippocampal Pattern Separation

Process of separating overlapping inputs into distinct memories to reduce interference in recall

✓ Testing Hypothesis

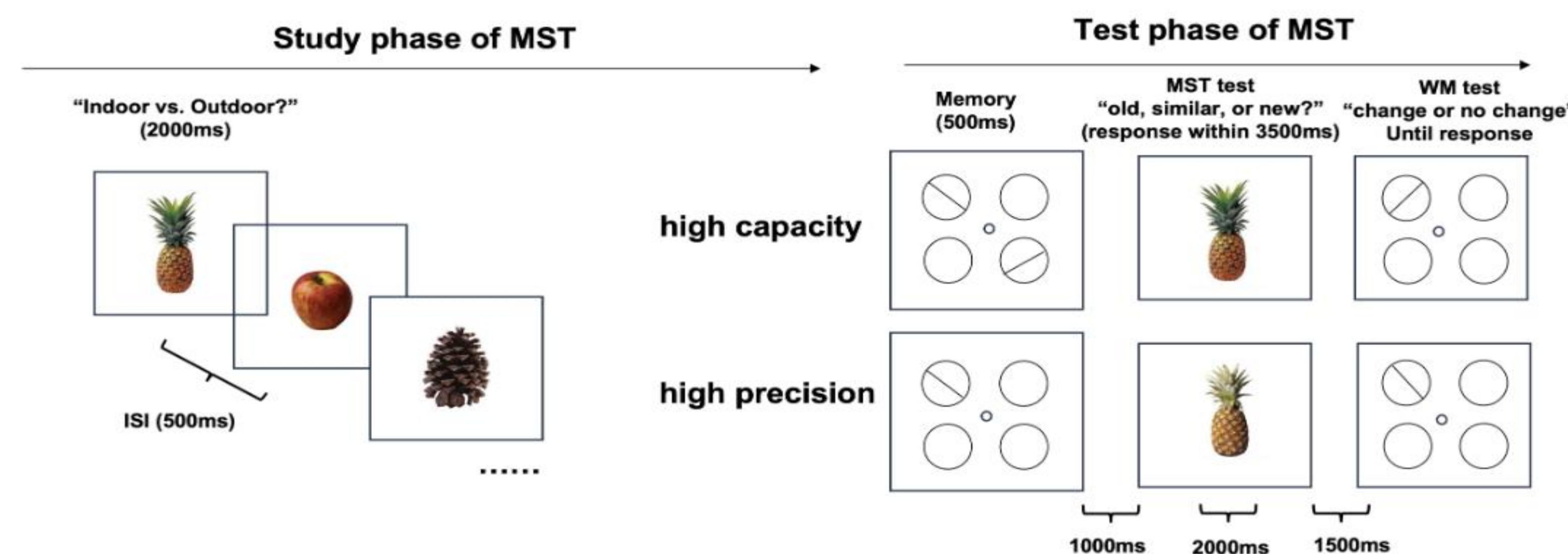
WM precision and hippocampal pattern separation share a common cognitive source and utilize a common cognitive source

✓ Prediction

High load on WM precision will reduce lure discrimination ability, but not on WM capacity.

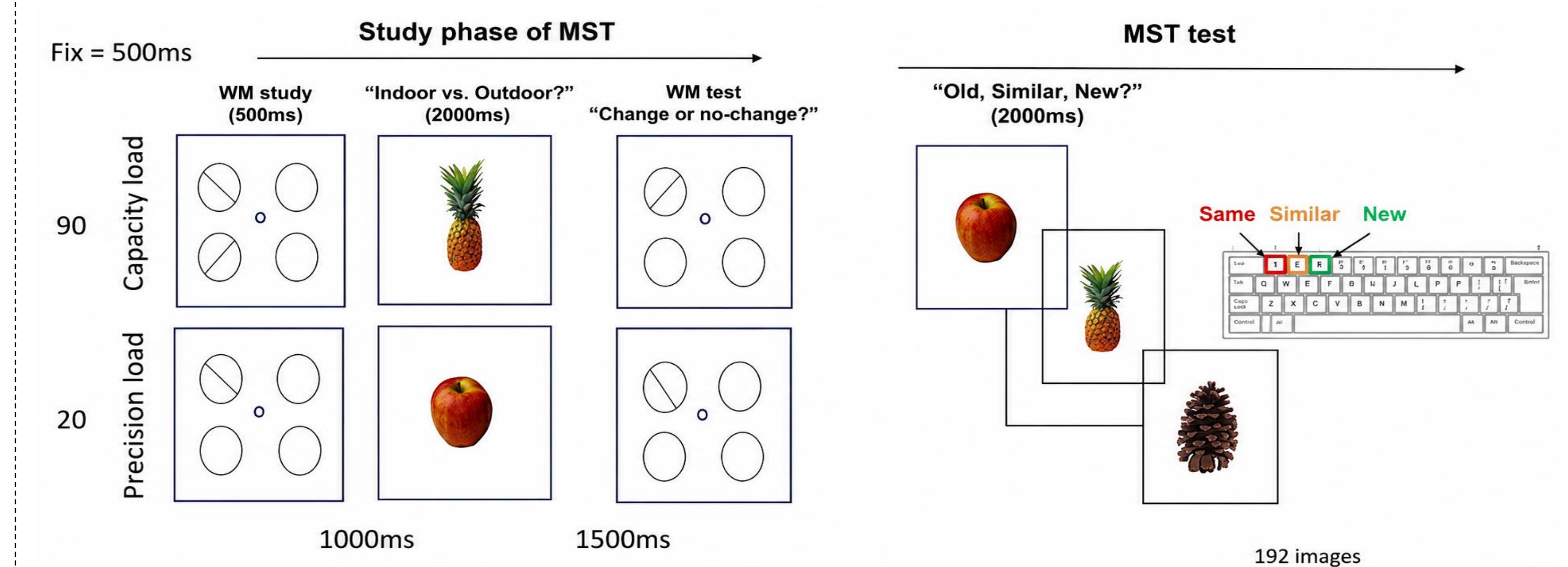
EXPERIMENT

✓ Experiment 1 Task and Procedure (N=53)



- The study phase of MST is for the **encoding** of a series of objects
- The test phase includes MST with **retrieval processes** with a memory test
- It is expected that **lure discrimination will decrease** during high WM precision load

✓ Experiment 2 Task and Procedure (N=45)



- The study phase includes MST with **encoding** process
- The test phase includes only the **retrieval** process
- It is expected that accuracy in WM **precision** load will be lower than in WM **capacity** load

DATA ANALYSIS

✓ Working Memory Accuracy

- Working memory (WM) accuracy measures how correctly participants memorize and respond to information during a memory task.
- It is calculated by dividing the number of correct responses by the total number of trials.
- WM accuracy was analyzed between the different MST trial conditions, including target, lure, and foil trials in Experiment 1 and target versus lure trials in Experiment 2.

✓ Experiment 1 Results

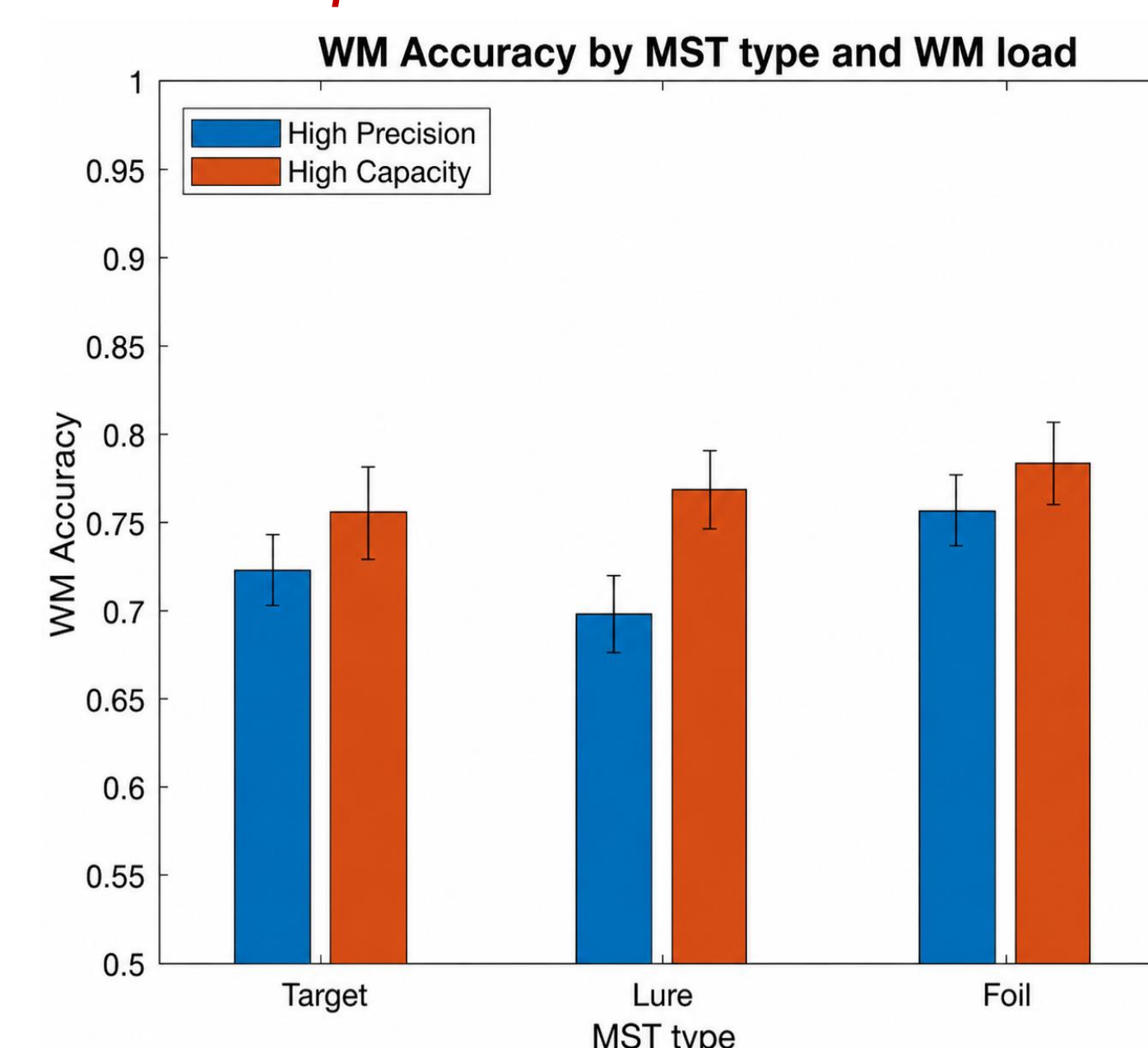


Figure 1. WM accuracy by MST type and precision vs capacity for Experiment 1

✓ Experiment 2 Results

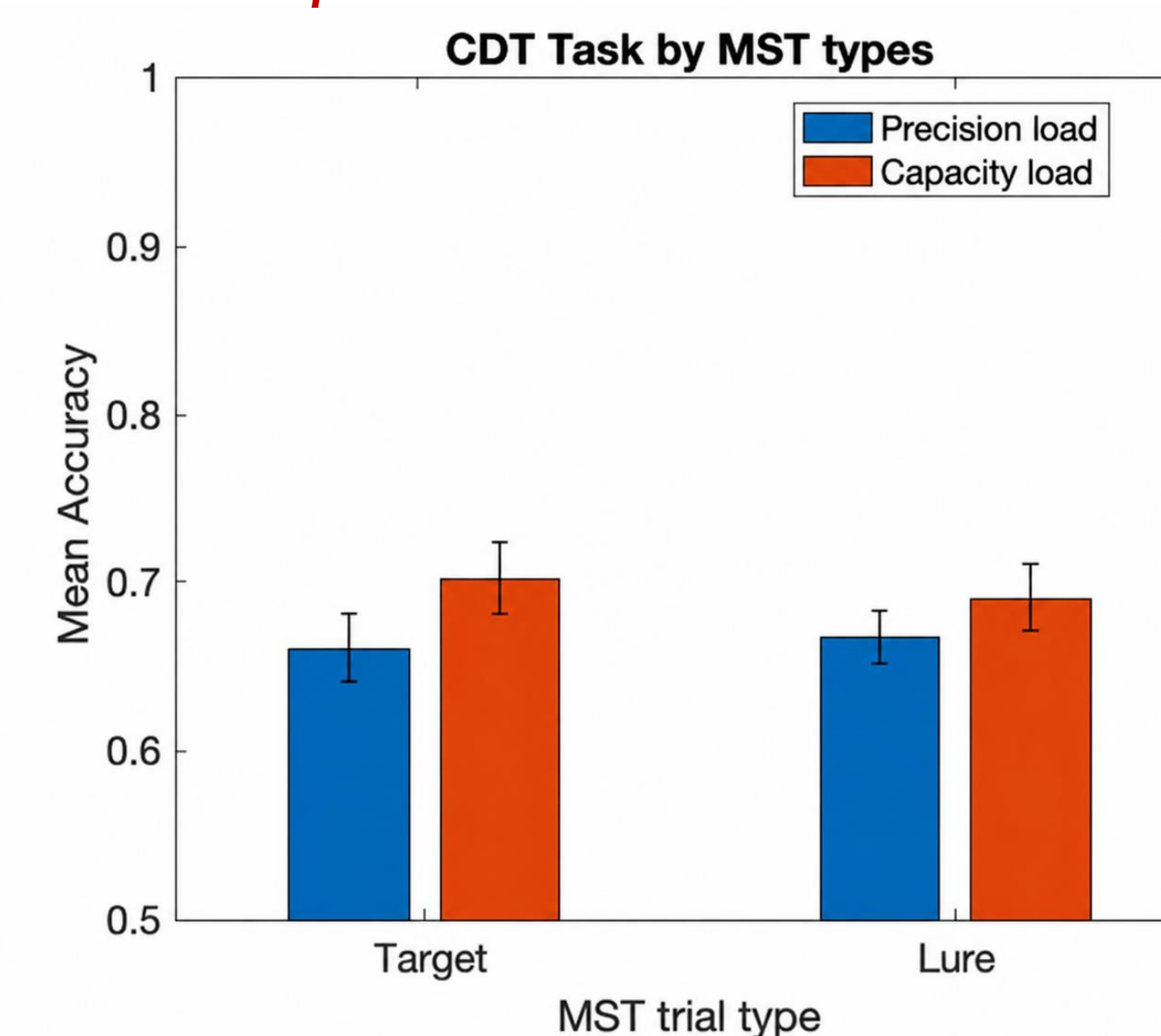


Figure 3. WM accuracy by MST type and precision vs capacity for Experiment 2

✓ LDI

- LDI (Lure Discrimination Index) measures how well participants distinguish similar "lure" items from previously seen items.
- LDI is calculated as $[p(\text{"similar"}|\text{lure}) - p(\text{"similar"}|\text{foil})]$

✓ Results

- There was no significant difference in LDI between WM loads for both experiments
- Only in the Experiment 1 Lure conditions is there a significant effect between loads

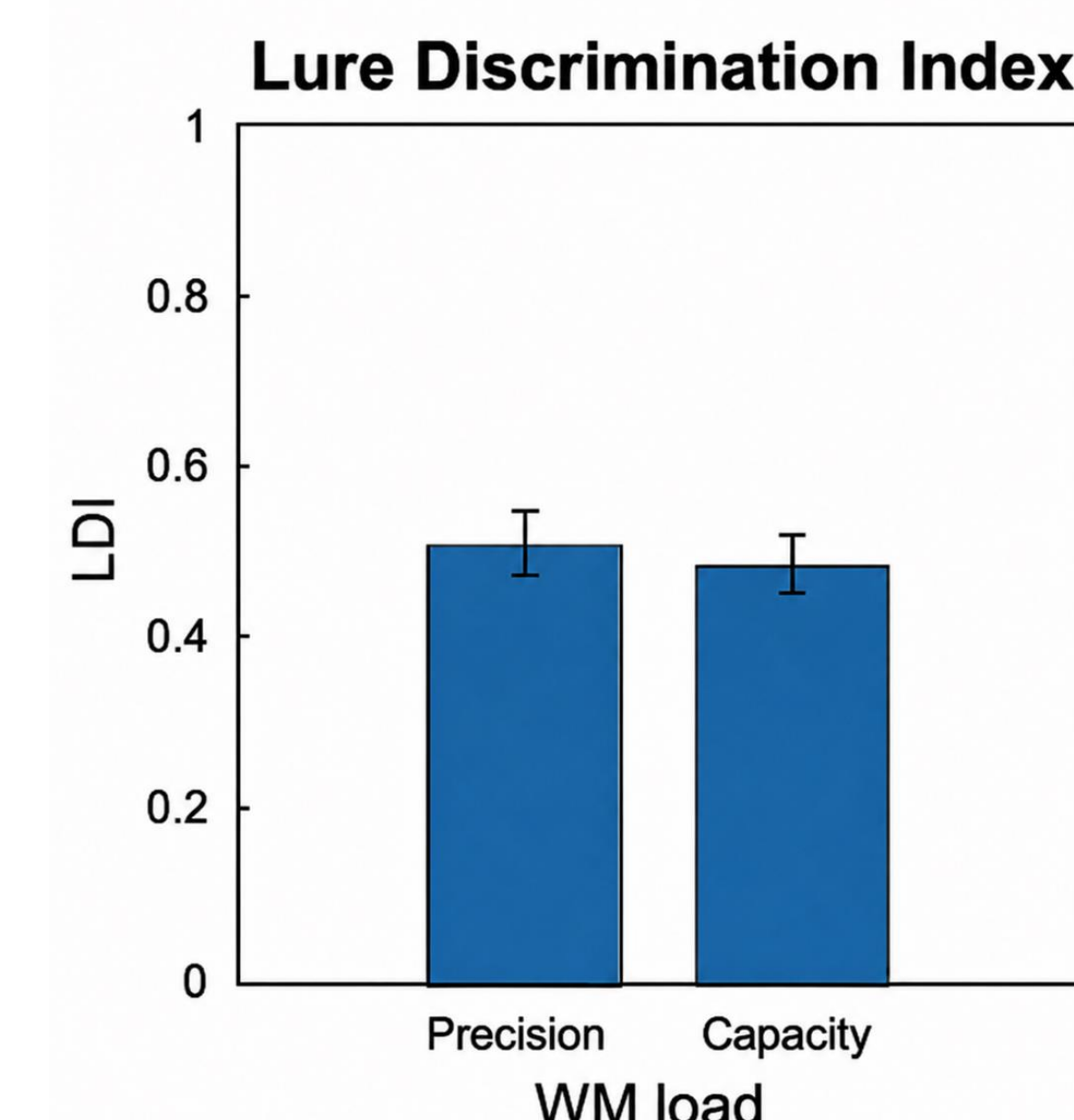


Figure 2. Lure Discrimination Index by WM load for Experiment 1

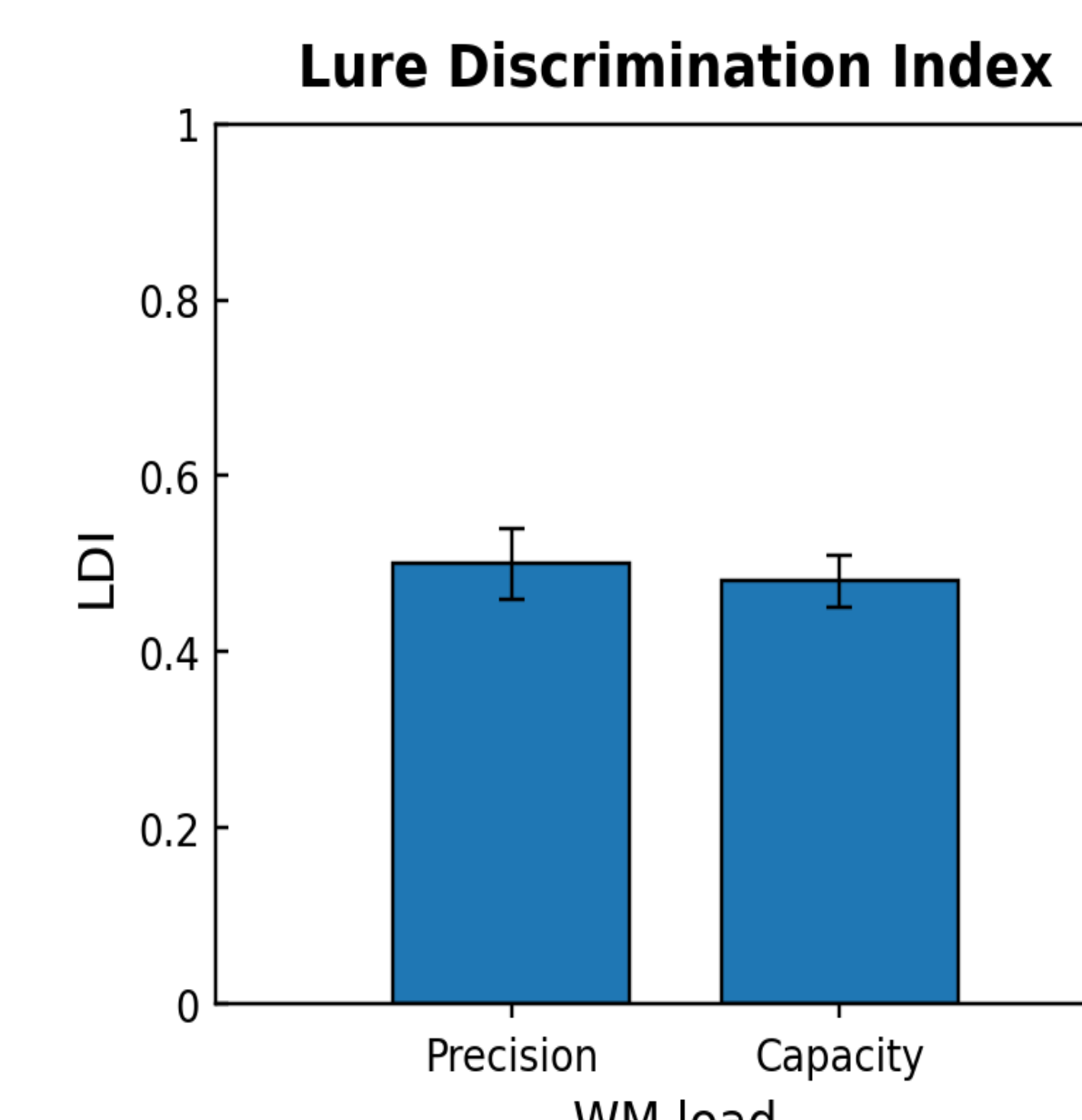


Figure 4. Lure Discrimination Index by WM load for Experiment 2

Discussion

- Competition between WM precision and hippocampal pattern separation during LTM retrieval
- Suggests that demands on LTM override precision, despite high loads

✓ Limitations & Implications

Formatting of MST focuses more on perceptual discrimination over WM precision. Can be overcome by isolating executive functioning from hippocampal pattern separation

Educational methods can be changed to teach students more effectively. Understanding the effects on working memory in relation to LTM and STM for diseases

References

1. Gellersen, H. M., Trelle, A. N., Henson, R. N., & Simons, J. S. (2021). Executive function and high ambiguity perceptual discrimination contribute to individual differences in mnemonic discrimination in older adults. *Cognition*, 209.
2. Xie, W., Cappiello, M., Yassa, M. A., Ester, E., Zaghoul, K. A., & Zhang, W. (2023). The entorhinal-DG/CA3 pathway in the medial temporal lobe retains visual working memory of a simple surface feature
3. Yassa, M. A., & Stark, C. E. L. (2011). Pattern separation in the hippocampus. In *Trends in Neurosciences* (Vol. 34, Issue 10, pp. 515–525)
4. Zhang, W., & Luck, S. J. (2015). Opposite effects of capacity load and resolution load on distractor processing. *Journal of Experimental Psychology: Human Perception and Performance*, 41(1), 22